B1.4 Conversion, protection, and detection

Connection of an AXI4-Lite slave to an AXI4 master requires some form of adaptation if it can not be ensured that the master only issues transactions that meet the AXI4-Lite requirements.

This section describes techniques that can be adopted in a system design to aid with the interoperability of components and the debugging of system design problems. These techniques are:

Conversion This requires the conversion of all transactions to a format that is compatible with the AXI4-Lite requirements.

Protection This requires the detection of any non-compliant transaction. The non-compliant transaction is discarded, and an error response is returned to the master that generated the transaction.

Detection This requires observing any transaction that falls outside the AXI4-Lite requirements and:

- notifying the controlling software of the unexpected access
- permitting the access to proceed at the hardware interface level.

B1.4.1 Conversion and protection levels

Different levels of conversion and protection can be implemented:

Full conversion

This converts all AXI transactions, as described in *Defined conversion mechanism* on page B1-125.

Simple conversion with protection

This propagates transactions that only require a simple conversion, but suppresses and error reports transactions that require a more complex task.

Examples of transactions that are propagated are the discarding of one or more of **AxLOCK** and **AxCACHE**.

Examples of transactions that are discarded and generate an error report are burst length or data width conversions.

Full protection

Suppress and generate an error for every transaction that does not comply with the AXI4-Lite requirements.

B1.4.2 Implementation considerations

A protection mechanism that discards transactions must provide a protocol-compliant error response to prevent deadlock. For example, in the full AXI protocol, read burst transactions require an error for each beat of the burst and a correctly asserted **RLAST** signal.

Using a combination of detection and conversion permits hardware implementations that:

- do not prevent unexpected accesses from occurring
- provide a mechanism for notifying the controlling software of the unexpected access, so speeding up the debug process.

In complex designs, the advantage of combining conversion and detection is that unforeseen future usage can be supported. For example, at design time it might be considered that only the processor programs the control register of a peripheral, but in practice, the peripheral might need to be programmed by other devices, for example a DSP or a DMA controller, that cannot generate exactly the required AXI4-Lite access.

The advantages and disadvantages of the different approaches are:

- Protection requires a lower gate count.
- Conversion ensures the interface can operate with unforeseen accesses.
- Conversion increases the portability of software from one system to another.

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- Conversion might provide more efficient use of the AXI infrastructure. For example, a burst of writes to a FIFO can be issued as a single burst, rather than needing to be issued as a set of single transactions.
- Conversion might provide more efficient use of narrow links, where the address and data payload signals are shared.
- Conversion might provide more flexibility in components that can be placed on AXI4-Lite interfaces. By converting bursts and permitting sparse strobes, memory can be placed on AXI4-Lite, with no burst conversion required in the memory device. This is, essentially, a sharing of the burst conversion logic.

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Part C

ACE Protocol Specification

Chapter C1
About ACE

This chapter gives an overview of system level coherency and the ACE protocol that supports it. It contains the following sections:

- *Coherency overview* on page C1-132
- *Protocol overview* on page C1-134
- *Channel overview* on page C1-137
- *Transaction overview* on page C1-142
- *Transaction processing* on page C1-146
- *Concepts required for the ACE specification* on page C1-147
- *Protocol errors* on page C1-150.

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